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PEE 414 — Past Questions & Materials
by Renance

Course: PEE 414
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 400

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Paper 1 of 3 — PEE 414 Petroleum Production Engineering I

Level 400 • First Semester • PastQuestion

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**FEDERAL UNIVERSITY OF PETROLEUM
RESOURCES, EFFURUN**



**COLLEGE OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF PETROLEUM ENGINEERING
2024/2025 First Semester Examinations**

PEE 414: Petroleum Production Engineering I

Answer any four Questions
Time Allowed: 3hrs Date: 16/05/2025

QUESTIONS

1a (i). What are the two key roles of the production technologist?
 (ii). What is 'perforation'? Will you consider it necessary to perforate a casing? Give a reason for your answer.

1b (i). State four (4) potential formation damage problems that can be experienced during production.
 (ii). A productivity test was conducted on a well. The test results indicate that the well is capable of producing at a stabilized flow rate of 100stb/d and a bottom-hole flowing pressure of 900psi. After shutting the well for 24hrs, the bottom-hole pressure reached a static value of 1200psi. Calculate;

- Productivity Index
- AOF
- Oil flow rate at a bottom-hole flowing pressure of 600psi.
- Wellbore flowing pressure required to produce 250stb/d.

2a (i). What is (are) the reason(s) for using propping agents during hydraulic fracturing
 (ii). What is IPR curve and what is its use in petroleum production engineering?

2b (i). What is (are) the assumption(s) considered in straight-line IPR? State its limitation(s) if any.

(ii). Using the following data;

- Porosity: $= 0.19$ *When the tubing size is 3.5 inch, friction loss is*
- Effective horizontal permeability: $= 8.2\text{md}$ *effective*
- Pay zone thickness: $= 53\text{ ft}$ *when the tubing size is 3.5 inch*
- Reservoir pressure: P_e or $= 5,651\text{ psia}$ *large additional pressure (psi)*
- Bubble-point pressure: $P_b = 50\text{ psia}$ *due to liquid loading*
- Fluid formation volume factor: $B_o = 1.4$
- Fluid viscosity: $= 1.7\text{ cp}$
- Total compressibility, $c_t = 0.0000129\text{ psi}^{-1}$
- Drainage area: $A = 640\text{ acres}$
- $r_e = 2,980\text{ ft}$
- $r_w = 0.328\text{ ft}$
- Skin factor: $S = 0$

Construct IPR of a vertical well in an oil reservoir considering

- steady-state flow, and $\rightarrow J < J_{KH}$
- pseudo-steady-state flow.

3a (i). What is Well Stimulation? Explain in detail, its purpose, objectives and uses.
 (ii). State any six (6) factors affecting VLP and explain three (3) of the factors.

3b (i). During completion planning design, how will the sizing of tubing affect a well system during production? What are the factors considered?

- 3b (ii). An E & P Co. operating in the Niger Delta, owns OMLxx well that is producing from a reservoir having an average pressure of 2200psig and is produced at a rate of 270 stb/d, with a bottom-hole pressure of 1700 psig. Calculate:

- The PI
- The producing rate, if Pwf is decreased to 1500 psig
- The FBHP necessary to increase the producing rate up to 600stb/d
- The inflow rate, if Pwf is reduced to zero (AOF)

- 4a (i). Sketch a typical pressure flow rate diagram to explain the pressure distribution behavior in a production system.

- (ii). State the similarities as well as dissimilarities between acidizing and fracturing.

- 4b (i). List the most common types of acid system used in stimulation operation.

- (ii). Construct IPR of a vertical well in an undersaturated oil reservoir using the generalized Vogel equation. The following data are given:

- Porosity: = 0.19
- Effective horizontal permeability: = 8.0 md
- Pay zone thickness: h = 53 ft
- Reservoir pressure: $\bar{P} = 5,650$ psia
- Bubble point pressure: $P_b = 3,000$ psia
- Fluid formation volume factor: $B_o = 1.2$
- Fluid viscosity: = 1.6 cp
- Total compressibility: = 0.000129 psi⁻¹
- Drainage area: A = 640 acres

$$\bullet r_e = 2,980 \text{ ft}$$

$$\bullet r_w = 0.328 \text{ ft}$$

$$\bullet \text{Skin factor: } S = 0$$

$$(1) J = \frac{q}{P - P_{wf}} = \frac{270}{2200 - 1700} = 0.54 \text{ STB/d-psi}$$

$$(2) q = J(P - P_{wf}) = 0.54(2200 - 1500) = 378 \text{ STB/d}$$

$$(3) q = J(\bar{P} - P_{wf}) = 0.54(5650 - 3000) = 1449 \text{ STB/d}$$

$$(4) q = J(\bar{P} - P_{wf}) = 0.54(5650 - 0) = 3051 \text{ STB/d}$$

- 5a (i). State the four major activities embarked on during well completion i.e. how are wells completed?

- (ii). Explain briefly in not more than two sentences, why wells are designed by the production engineer?

- 5b (i). What do you understand by the term "Drill Stem Testing"?

- (ii). A well is producing from an undersaturated oil reservoir that exists at an average reservoir pressure of 4000 psi. The bubble point pressure is recorded as 2200 psi at 150°F, the following additional data are available:

- Stabilized wellbore pressure = 2500 psi
- Pay-zone thickness = 6.096 m
- Well radius = 0.3 ft.
- Drainage radius = 20117 cm
- Skin due to perforation, $S_p = 4$
- Permeability = 65 md
- Oil viscosity = 2.4 cp
- Oil formation volume factor = 1.4 bbl/STB

Calculate the flow rate, q (STB/day) at wellbore pressures of 1000, 1800, 2000 and 3000 psi.



Paper 2 of 3 — PEE 414 PETROLEUM PRODUCTION ENGINEERING I 2024 2025

Level 400 • First Semester • PastQuestion

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- Fluid viscosity: = 1.6 cp
- Total compressibility: = 0.0000129 psi^{-1}
- Drainage area: $A = 640$ acres

- $r_e = 2,980$ ft
- $r_w = 0.328$ ft
- Skin factor: $S = 0$

$P - P_{wf} = 1700 - 100$
 $q = J (P - P_{wf}) = 0.3333 (1700 - 100)$
 $= 199.98 \text{ STB/d}$
 $q = J (P - P_{wf})$
 $250 = 0.3333 (1200 - P_{wf})$
 $P_{wf} = 449.92$

- 5a (i). State the four major activities embarked on during well completion i.e. how are wells completed?
- (ii). Explain briefly in not more than two sentences, why wells are designed by the production engineer?

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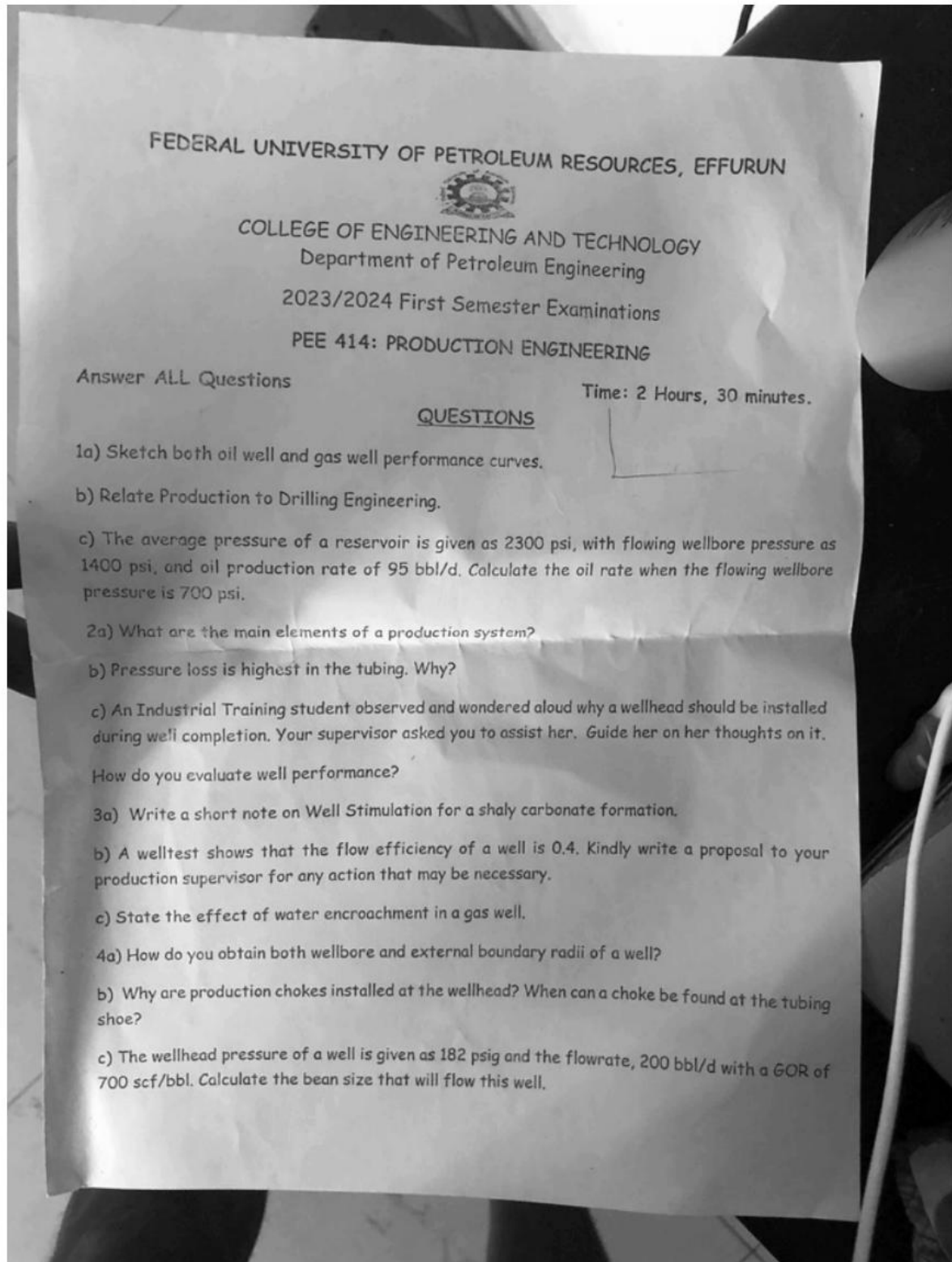
Calculate the flow rate, q (STB/day) at wellbore pressures of 1600, 1800, 2000 and 3000 psi.



Paper 3 of 3 — PEE 414 PQ 2024

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4bi) Mention the devices used for manual gauging and sampling operation.

4c) why do you isolate the tank in tank venting and cleaning

Question five

5a) what is the processes of returning the tank to source after maintenance or repairs?

5ai) List and explain 4 fire protection and prevention

5aii) Distinguish between flow line gathering lines and pipelines in a marginal field.

5b) what is the EPA recommendation for flow line maintenance.

Question six

6a) what does the production characteristics of each well depends on?

6ai) Mention the 4 most popular way of transporting crude oil and gas all over the world.

6b) In not more than 2 sentences explain the requirement of personnel who load and offload Hydrocarbon.